

BIO101

BIODIVERSITY OF ANIMALS PLANTS & MICROBES

AY2026-27



DR. SEKAR RAJU



Xi'an Jiaotong-Liverpool University

西交利物浦大學

BIO101 – General Information 1

- **Module Leader**

Dr Sekar Raju

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Email: Sekar.Raju@xjtlu.edu.cn

- **Credits / Level**

5 credits; Level 1 (Year 2 / Stage 2)

- **Lecture Time and Venue:**

Monday 4:00 PM to 6:00 PM (FB296)

Wednesday 12:00 to 1:00 PM (MA104)

- **Tutorials:** **Week 5, 8, 11 Friday; 11 AM to 1:00 PM (SC140)**
ALC Tutorial: **Week 4, 9, 12 Friday; 11 AM to 1:00 PM (SC140)**
- **Office Hours:** **Friday 2:00 PM – 4:00 PM (SA443)**



BIO101 – General Information 2



- **Academic Literacies Centre (ALC)**
Support Staff for BIO101



Ms. Xinyi Chen

Office: FB439

Email: Xinyi.Chen@xjtlu.edu.cn

Office hours: Wednesday 2 to 4 PM

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- **Support provided by the ALC**

🔍 Vocabulary building

✍️ Writing Skills

👥 Workshops/Consults



💪 Confidence Building

📚 Integrated Skills

📊 Academic Success

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Module Handbook: Available on eBridge and Learning Mall Core

Module Handbook

Module(s)						
Academic Year~Semester	Module Code	Module Title	Credit	Status	Action	Current Version
2026/27~SEM1	BIO101	Biodiversity of Animals, Plants and Microbes	5	Submitted and Released to Students	RE-OPEN TO EDIT	VIEW



[MODULE HANDBOOK SYNCHRONIZATION TO LMO](#)

[DOWNLOAD PDF VERSION](#)

BIO101

Biodiversity of Animals, Plants and Microbes
Biological and Biomedical Sciences Stage 2 | Level 1

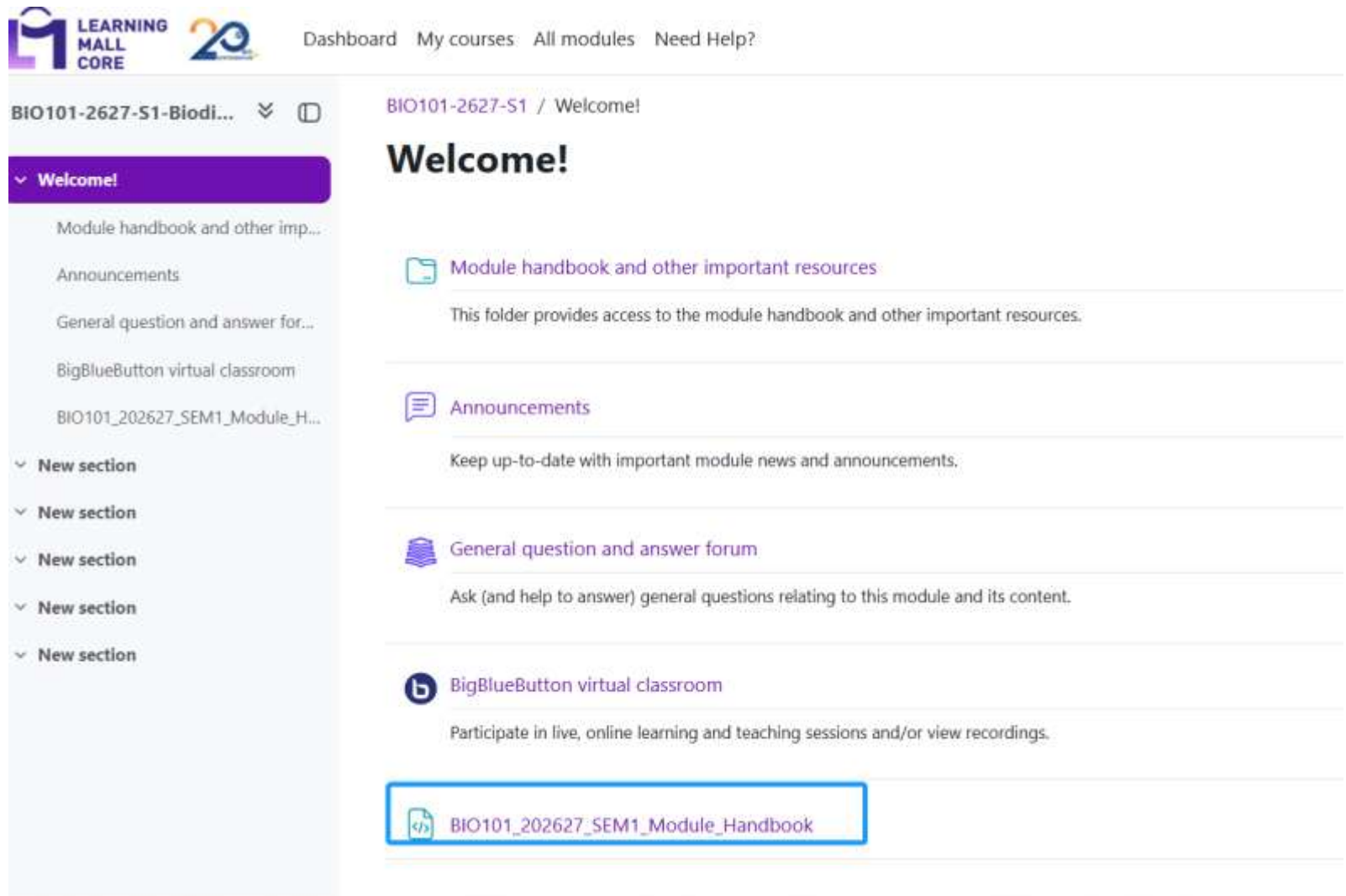
SECTION A: Basic Information

Brief Introduction to the Module

This module tackles the subject of biological diversity by combining systematic and functional analysis of the major groups of organisms responsible for sustaining life on earth. The origins of life, evolution of the major domains and an examination of microbial diversity lead into studies on plant communities, their underlying biology and exploitation by man. This module also enables students to gain an understanding of the structure and function of the basic body plan of three groups of animals. These all hold key positions in the evolutionary development of animal life. The three groups are: Echinoderms, which have embryological affinities with the vertebrates; Insects, which are numerically and economically the most varied and important group of invertebrates; and mammals, which are considered to be the most advanced; and the most intelligent of the vertebrate groups.

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Module Handbook: Available on Learning Mall Core



LEARNING MALL CORE  Dashboard My courses All modules Need Help?

BIO101-2627-S1-Biodi...  

Welcome!

- Module handbook and other imp...
- Announcements
- General question and answer for...
- BigBlueButton virtual classroom
- BIO101_202627_SEM1_Module_H...

New section

New section

New section

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New section

BIO101-2627-S1 / Welcome!

Welcome!

-  **Module handbook and other important resources**
This folder provides access to the module handbook and other important resources.
-  **Announcements**
Keep up-to-date with important module news and announcements.
-  **General question and answer forum**
Ask (and help to answer) general questions relating to this module and its content.
-  **BigBlueButton virtual classroom**
Participate in live, online learning and teaching sessions and/or view recordings.
-  **BIO101_202627_SEM1_Module_Handbook**

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Attendance Management System (AMS)

<https://ams.xjtlu.edu.cn>

Sign-in Activity 

 QR Code

 Password

Module Code

BIO101

Module Title

Biodiversity of Animals, Plants and Microbes

Date

2026-09-07

Time

16:00~18:00 Please ensure that classes finish approximately ten minutes before the scheduled finish time for the venue to be vacated and for staff and students to move between venues.

Location

FB296

Teaching Group

BIO101-Lecture-D1/1

0
Present

136
Total

My module

My authorized module

2026-09-07



BIO101 Biodiversity of Animals, Plants and Microbes
2026-09-07
16:00~18:00
FB296

2026-09-09



BIO101 Biodiversity of Animals, Plants and Microbes
2026-09-09
12:00~13:00
MA104

Source: XJTLU User Guide for Attendance Management System

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	Topics Covered
Microbes	1) The Origin of Life; Introduction to Biological Diversity 2) Phylogeny; Determining the Phylogenetic Relationships; Microbial Evolution 3) Diversity of Bacteria, Archaea, Fungi and Viruses 4) The Importance of Bacterial Taxonomy (Medical and Biotechnological significance) 5) Microbial Interactions 6) The Protists; Importance of Protozoans (Medical and Geochemical Significance)
Plants	7) Classification and Plant Biodiversity 8) Seedless and Seed plants and their importance 9) Flowering Plants- Form, Function and Transport 10) The Crop Plants
Animals	11) Introduction to Animal Diversity; Size, Shape and Body Symmetry 12) Characteristics of Arthropoda 13) Sense Organs, Respiratory, Digestive and Excretory Systems in Insects and Mammals 14) Locomotion and Reproduction in Insects and Mammals 15) Echinoderms; Evolutionary Origins and Diversity of Echinoderms. 16) Origin and Evolution of Mammals 17) The Placental Mammals 18) The Primates and Human Evolution

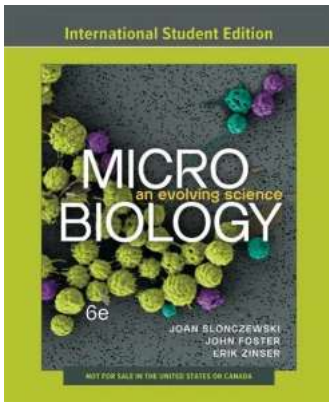
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Learning Outcomes:

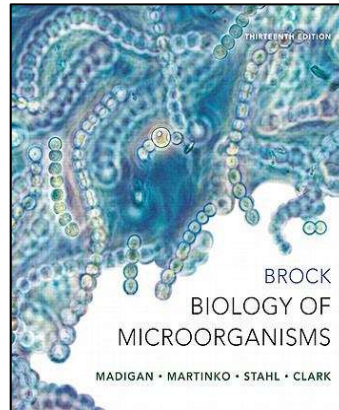
- The hypothesis for life's origins and evolution that defines biodiversity.
- Environmental importance of microorganisms and plants and how they impinge upon man's activities.
- Biological function in relation to the evolution of species.
- Structure and function of basic body plans of selected organisms e.g. insects, echinoderms and mammals.

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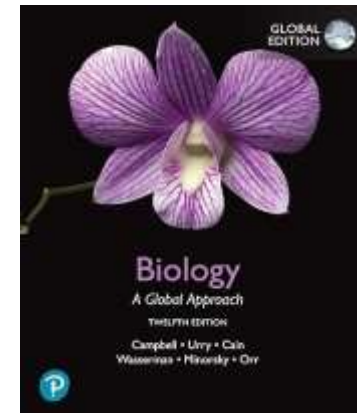
Mandatory, Optional and Reference Textbooks*



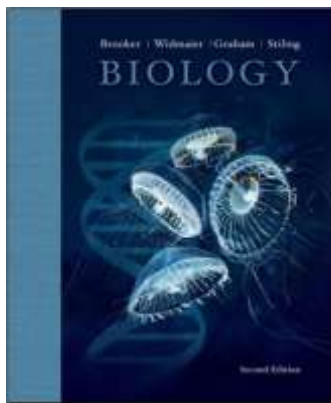
J. L Slonczewski et al.



Madigan et al.



Campbell et al.



Brooker et al.



Biology of Plants – Raven et al.

* Current or previous editions can be used

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Assessment

Initial Assessment

Sequence	Assessment Type(EXAM or CW)	Method	Learning outcomes assessed(<i>use codes under Learning Outcomes</i>)	Duration ⁴	% of Final Mark	Resit(Y/N/S) ⁵
001	CW	Quiz #1	A, B		15	S
002	CW	Quiz #2	B, C		10	S
003	CW	Quiz #3	D, E, F		15	S
004	EXAM	Final Exam	ALL	2.0 hours	60	S

Resit Assessment

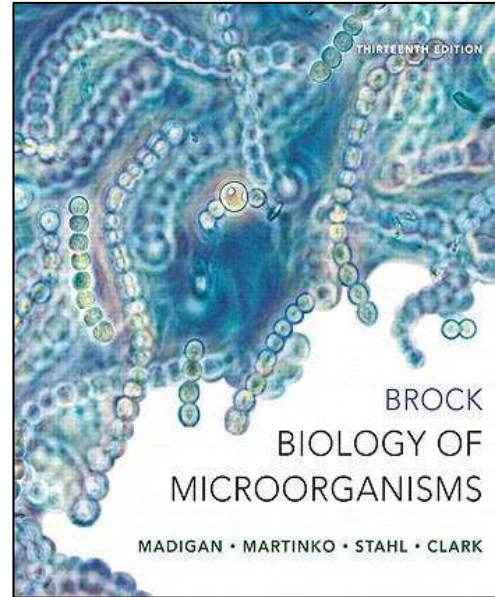
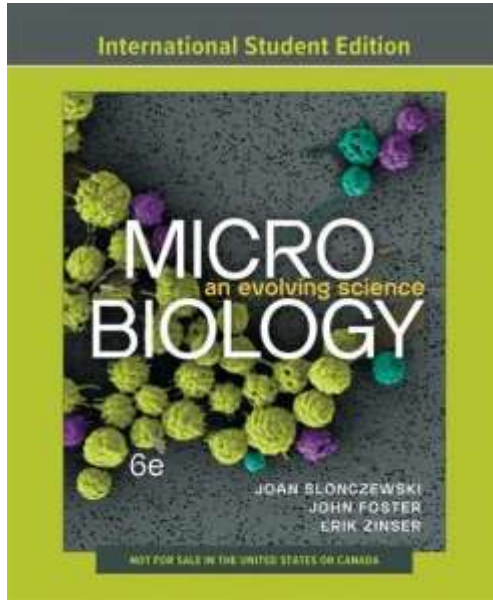
Assessment Type (EXAM or CW) ⁶	Learning outcomes assessed (<i>use codes under Learning Outcomes</i>)	Duration	% of Final Mark
EXAM	ALL	2.0 hours	100

The resit assessment will assess all of the learning outcomes of the module, and will be weighted as 100% of the final module mark. Other components of the assessment, regardless of whether or not the student passed or failed, will not be included in the calculation of the final module mark, following resit assessment.

Note: Quiz #1 and #3 will each include questions worth 5% from the ALC Team focusing on the English language.

Lecture 01

Origin of Life and Introduction to Biodiversity

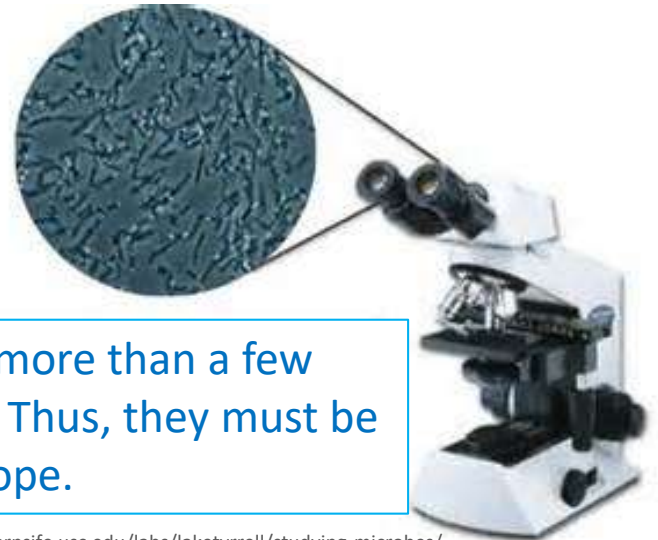


Refer to relevant chapters in the textbooks

Origin of life

Microbes came first

- A microbe is a living organism; can only be seen through a microscope. Examples: Bacteria, Fungi, Viruses and Protozoa.
- Most microbes consist of a single cell, but some are multicellular.
- Size varies: 0.2 micrometer (μm) to few millimeter (mm); Viruses are smaller than the smallest cells.



Most microbes are no more than a few micrometers in length. Thus, they must be viewed with a microscope.

Origin of life

Microscopic organisms

- Microbes include many different types of organisms.
 - **Prokaryotes** (cells lacking a nucleus): bacteria, archaea
 - Some **eukaryotes** (cells with a nucleus): algae, fungi, protists
 - **Viruses and prions*** (acellular entities)
- The bacteria, archaea, and eukaryotes—known as the three “**domains of life**”— evolved from a common ancestral cell.

** Prions - proteinaceous infectious particles*

Origin of life

Six major groups of microorganisms

Prokaryotes



Bacteria



Archaea

Eukaryotes



Algae

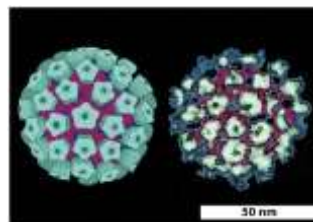


Protists



Fungi

Viruses



Origin of life

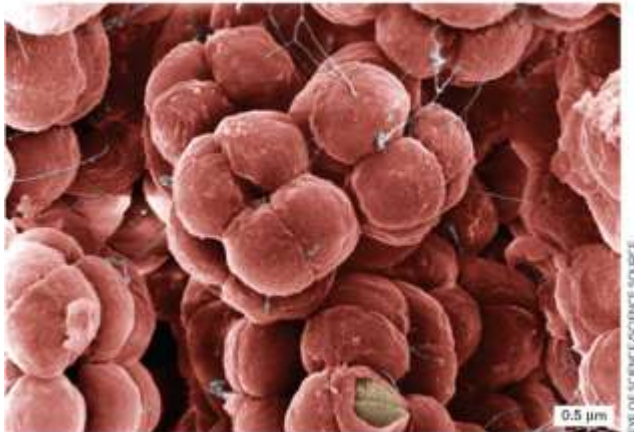
A. Bacteria: *E. coli*



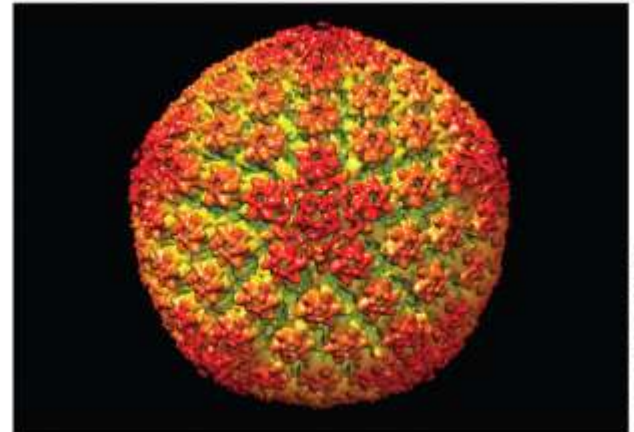
B. Eukaryote: *Stentor*



C. Archaea: Halophiles



D. Virus: Herpes simplex virus



A. Bacterium: *Escherichia coli* on leaf stomate. **B. Eukaryote:** *Stentor*, a stalked protist.
C. Archaeon: *Halococcus*. **D. Virus:** Herpes simplex virus.

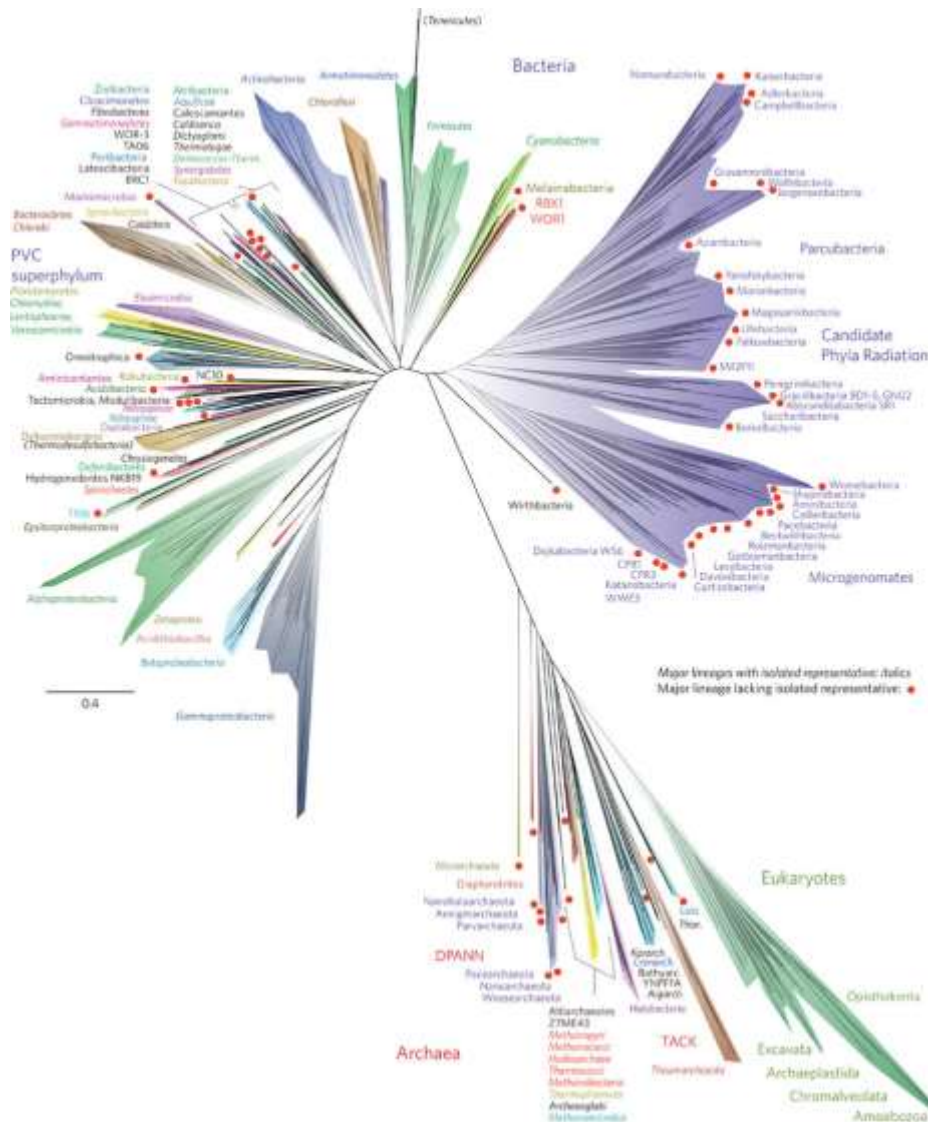
Origin of life

The Tree of Life

- Three domains
 - Bacteria
 - Archaea
 - Eukaryotes
- Most of life is microbial
- Microbes came first

Mammals

Hug, L.A. *et al.* 2016. A new view of the tree of life. *Nature Microbiol.* 1
doi: 10.1038/nmicrobiol.2016.48



Origin of life

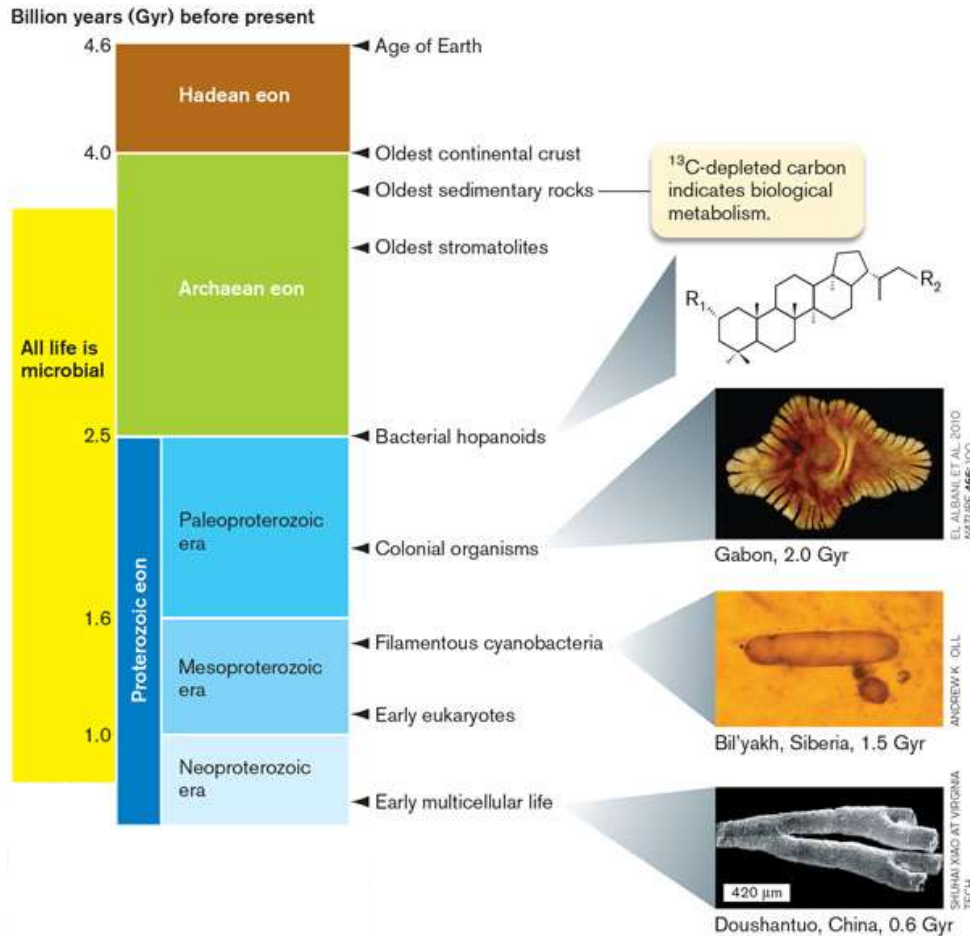
Microbes shape human history

- Microbes shaped human history since our earliest civilizations
- Microbes affect food availability
 - Destroy crops, but preserve food
 - Bread, wine, cheese
 - Variety of other fermented products
- Microbial diseases change history
 - Microbes caused diseases and famines
 - 'Black Death' (plague) in Europe
 - Smallpox in Americas



Origin of life

Geological Evidence for Early Life



- Evidence of microbial life includes **isotope ratios, stromatolites, and microfossils.**
- The geological record shows biosignatures of microbial life early in Earth's history, 3 Gyr before the first multicellular forms.

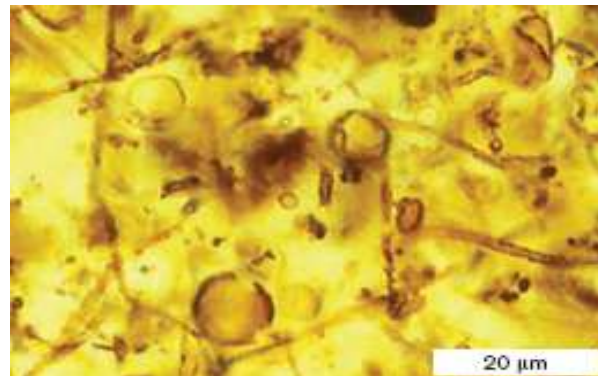
Origin of life

Microbes were early life forms on Earth

- First evidence for microbial life can be found in rocks about 3.86 billion years old.
- **Stromatolites:** Fossilized microbial mats consisting of layers of filamentous prokaryotes and trapped sediment. The sedimentary layers are limestone (calcium carbonate, CaCO_3) accreted by microbes over years or even centuries.
- **Microfossils:** small remains of organisms, in the range of 0.001mm (1 micron), invisible to naked eye e.g. fossils of bacteria, diatoms.



Stromatolites



Microfossils

Origin of life

Ancient stromatolites

- Anoxygenic phototrophic filamentous bacteria formed ancient stromatolites

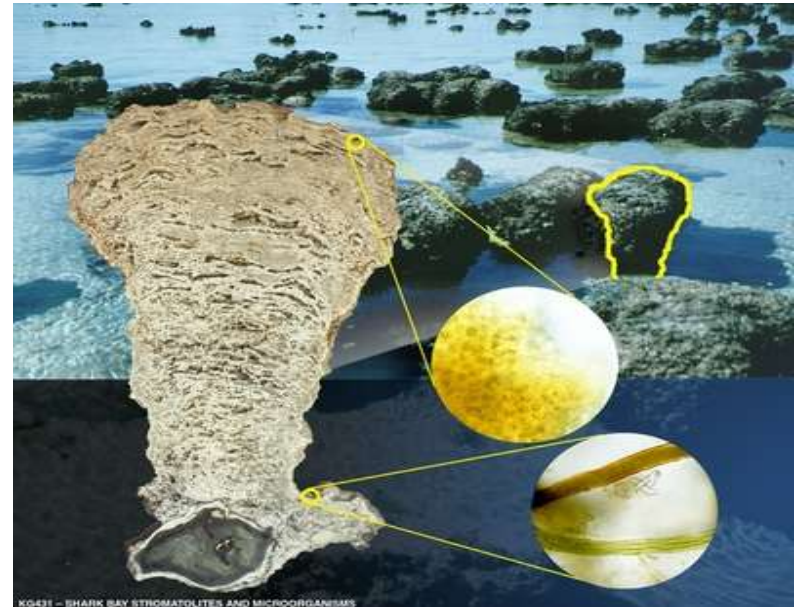


Pre-Cambrian stromatolites, Glacier National Park (Montana, USA)

Origin of life

Modern stromatolites

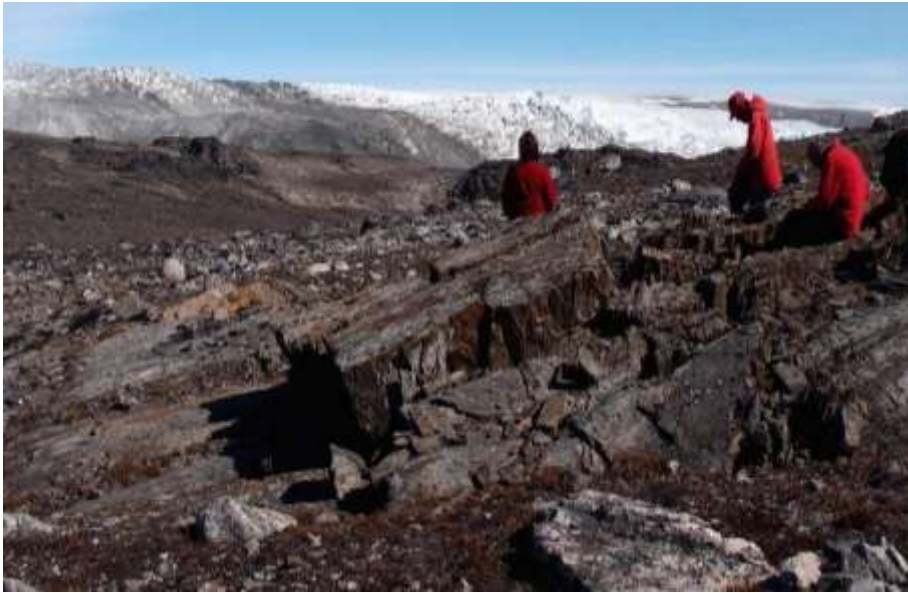
- Oxygenic (oxygen-producing) phototrophic cyanobacteria dominate modern stromatolites



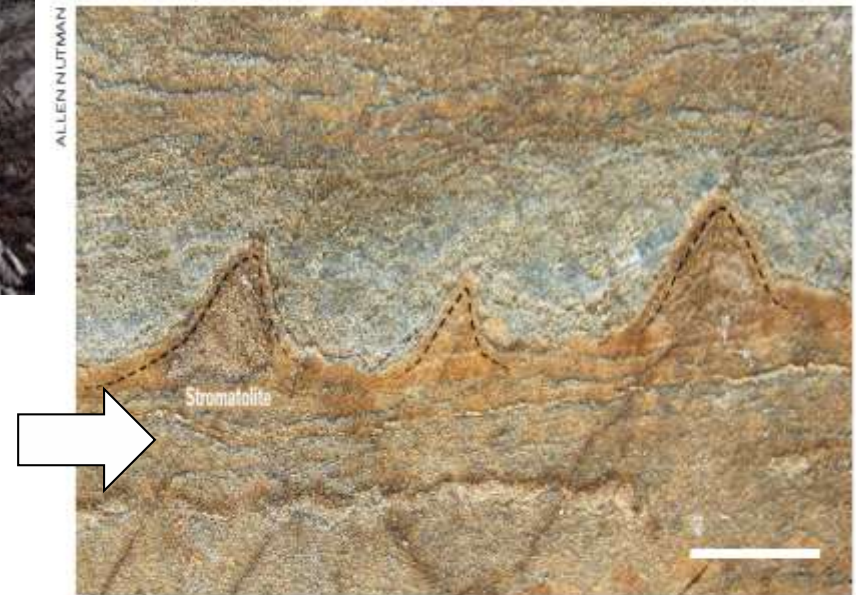
Modern stromatolites in Shark Bay, Western Australia

Origin of life

A finding in the last decade - the oldest evidence of life on Earth found in Western Australia*



A 3.7 billion year old rock with signs of associated life. Conical stromatolite-like structures are sign of bacterial activity.



* Nutman, A. P., et al. 2016. *Nature*
<http://dx.doi.org/10.1038/nature19355>

Origin of life

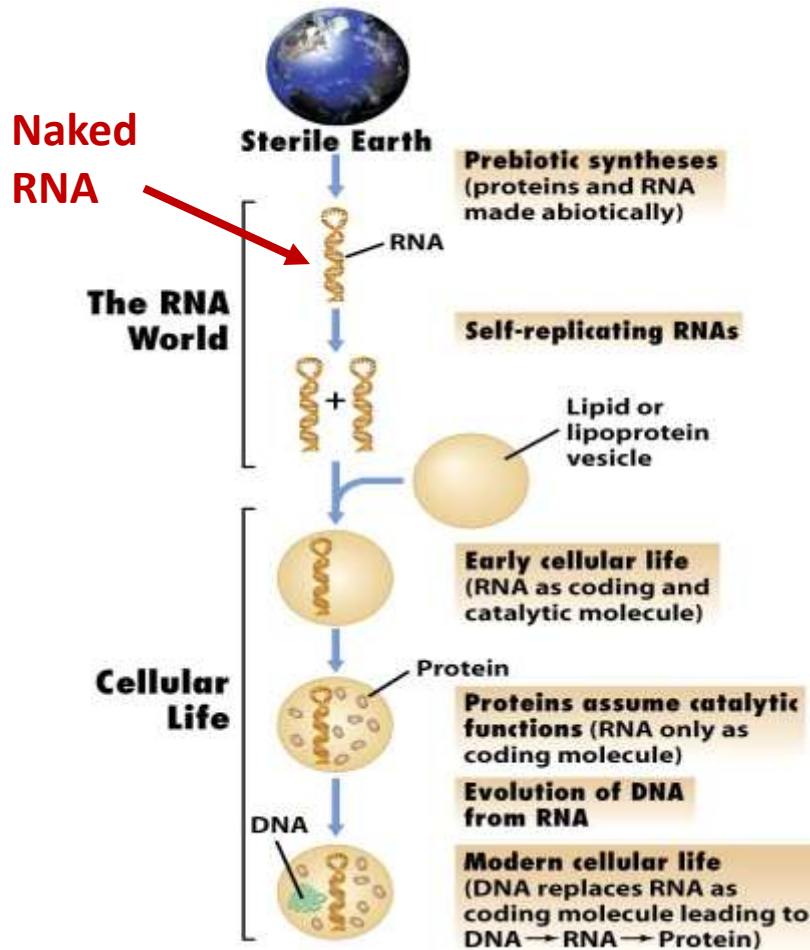
Conditions on early Earth: Hot and anoxic

- The atmosphere of early earth was devoid of oxygen
- Early earth was hotter than it is today
- Besides water, various gases were present:
 - Abundant - Methane, carbon dioxide, nitrogen & ammonia
 - Trace amounts – Carbon monoxide, hydrogen, H_2S (& FeS)
- Considerable amount of HCN was also produced on early Earth due to reaction of NH_3 with CH_4



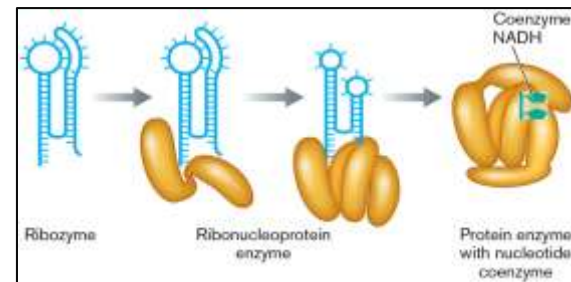
Origin of life

Primitive Life: The RNA world and molecular cloning



- First life forms probably self-replicating RNAs ('RNA life'). These were both catalytic and informational.
- Eventually, information store was DNA.
- Three-part system - DNA, RNA, and protein - became universal among cells.

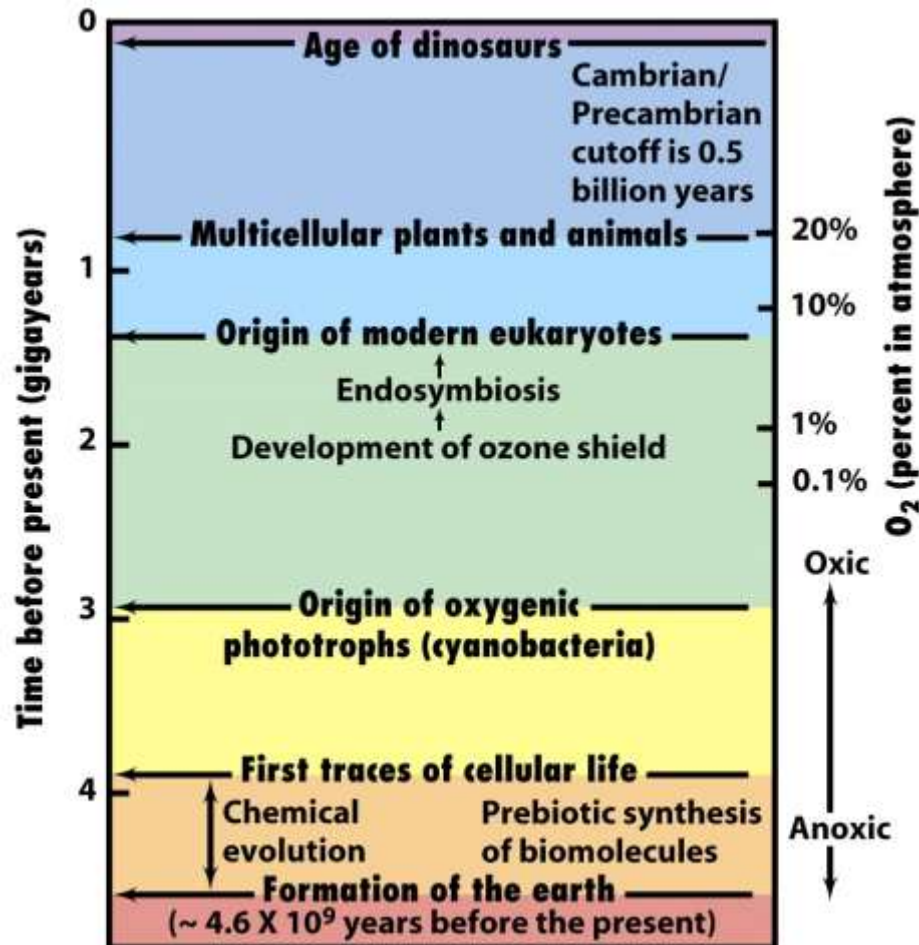
From the RNA world to proteins



The earliest cells may have been composed of RNA enzymes (ribozymes). As the RNA cells evolved, ribozymes acquired protein subunits that eventually assumed most of their catalytic functions. Remnants of the original RNA may persist as nucleotide cofactors such as NADH.

Origin of life

Oxygenation of the atmosphere

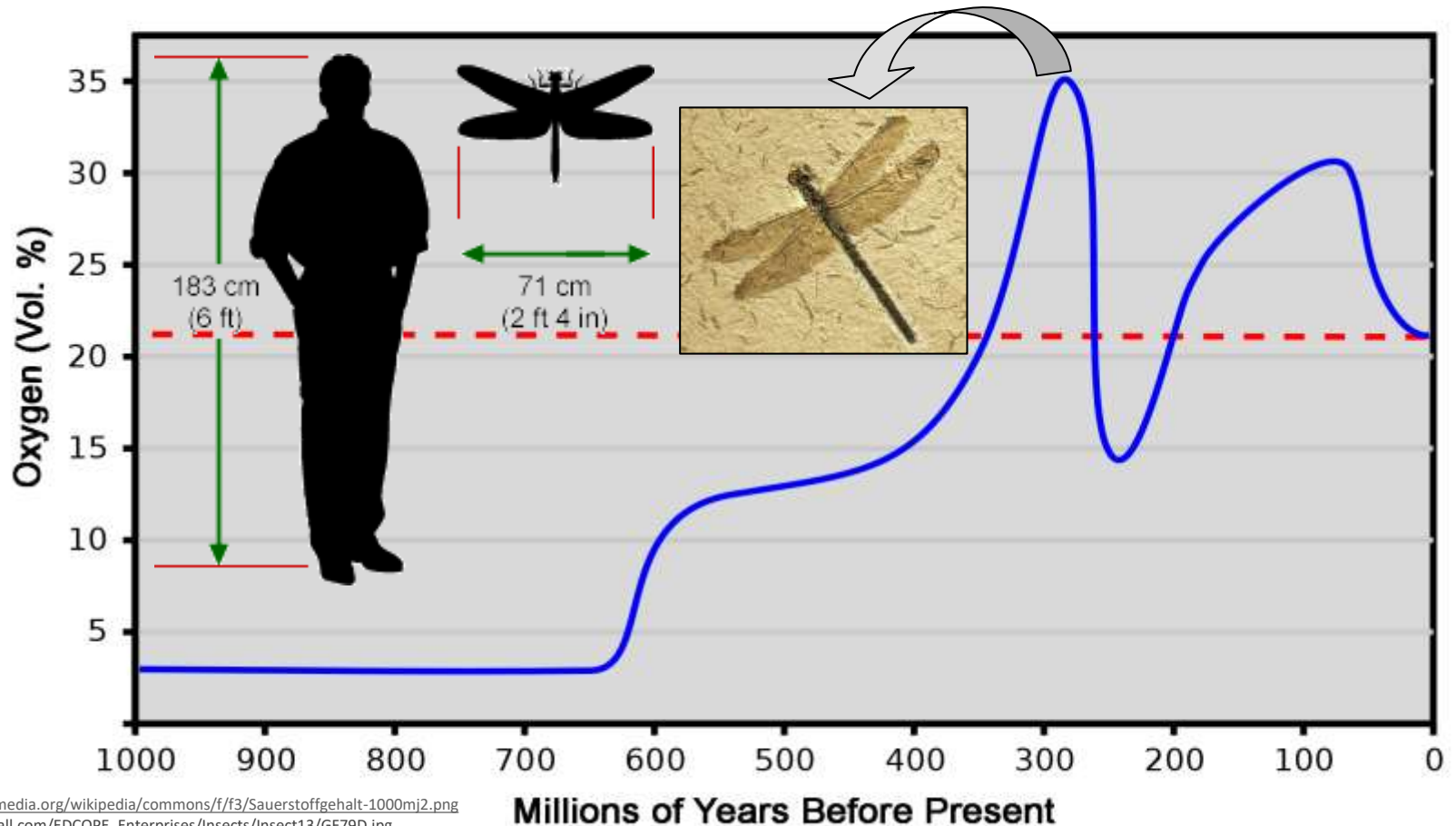


- Oxygenic (oxygen producing) photosynthesis led to an O₂-rich environment and great bursts of biological evolution
- Easy to spot in rocks - development of banded iron formations
- O₂-producing cyanobacteria appeared 2.8 billion years ago

Origin of life

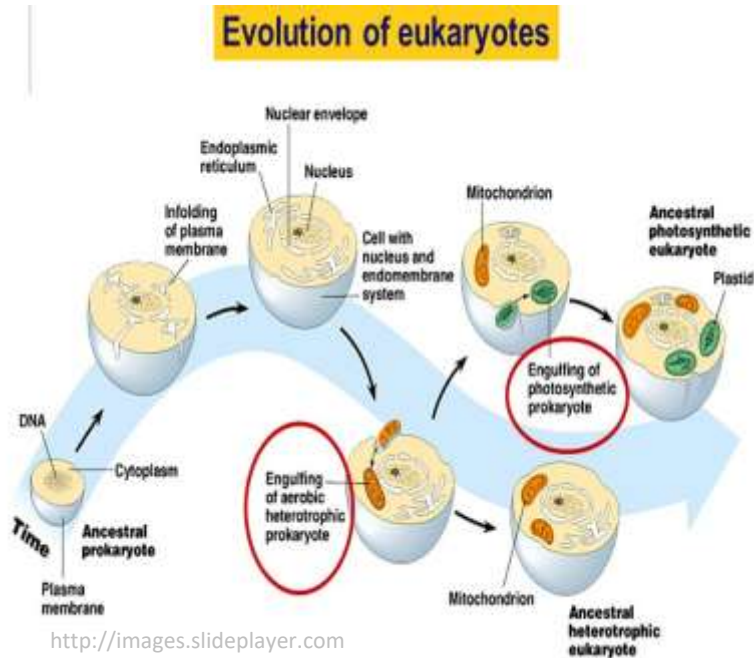
Oxygen Content of Earth's Atmosphere

During the Course of the Last Billion Years

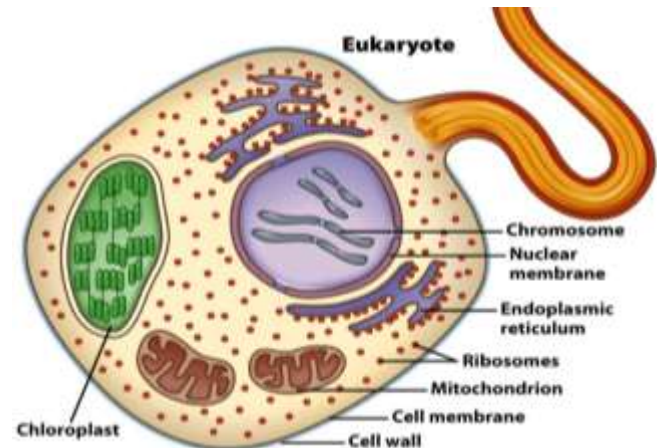


Origin of life

Eukaryotes and organelles - endosymbiosis



- Well-supported theory for origin of eukaryotic cells
- Mitochondria and chloroplasts, the principal energy-producing **organelles** of eukaryotes, arose when a prokaryote of the **domain *Bacteria*** entered another cell. This process is called **endosymbiosis**



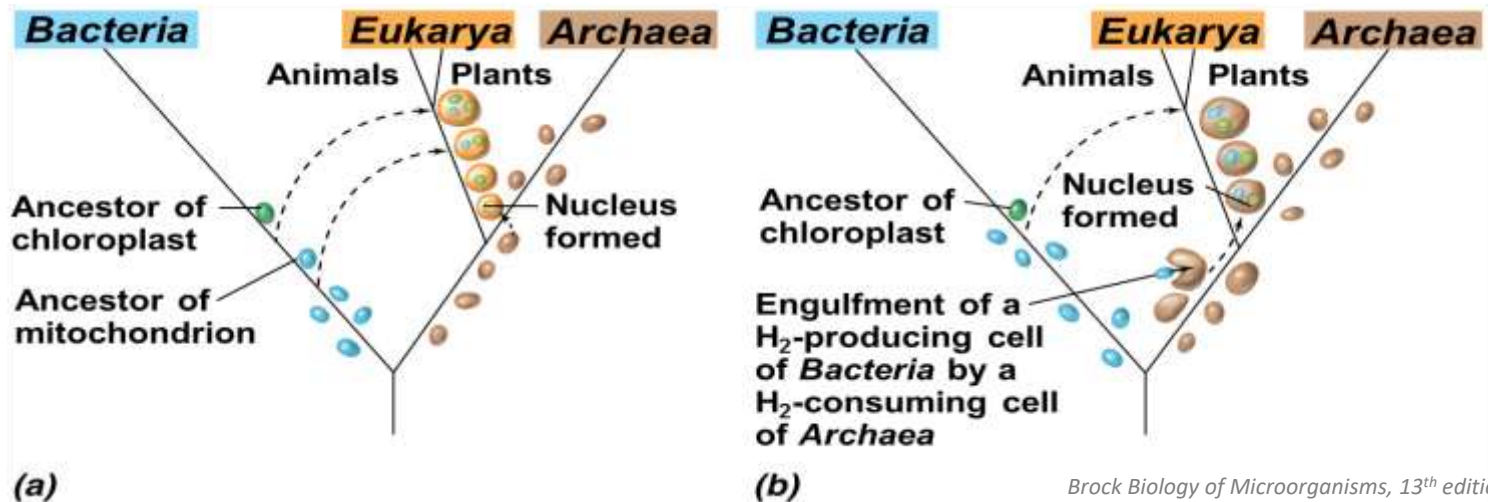
Origin of life

Origin of Eukaryotes – Two hypotheses

- a) Eukaryotes first arose from cell with a nucleus that later acquired mitochondria and chloroplasts by endosymbiosis .

OR

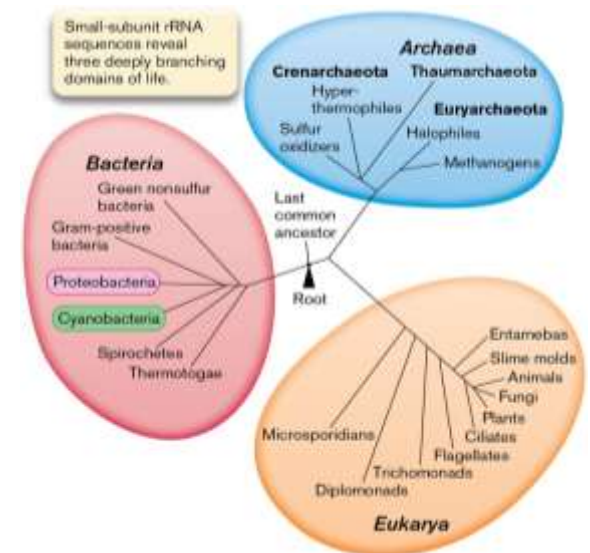
- b) Eukaryotes arose from intracellular association between O_2 -consuming bacterium (the symbiont), which gave rise to mitochondria, and an archaeal host.



Introduction to Biodiversity

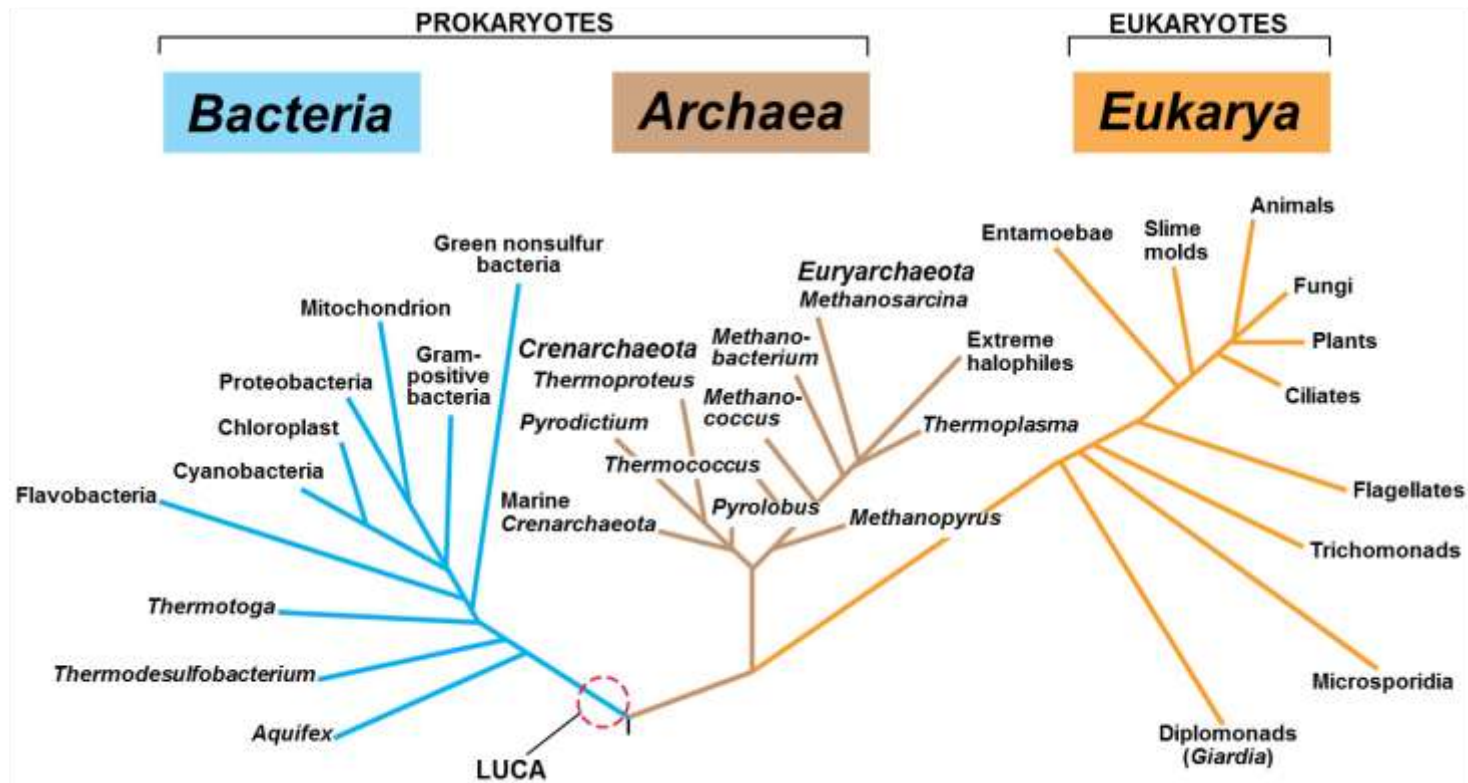
Three domains of life

- Three phylogenetically distinct lineages (domains) of cells have been identified:
 - Bacteria
 - Archaea
 - Eukaryotes
- The three domains are thought to have diverged from a common ancestral organism



Introduction to Biodiversity

Three domains of life



- Prokaryotes are not necessarily closely related phylogenetically.
- Archaea are more closely related to Eukaryotes than Bacteria.
- LUCA = Last Universal Common Ancestor.

Introduction to Biodiversity

Microbial diversity

- Microbial diversity we see now is ~4 billion years of evolutionary change
- Microbial diversity can be seen in many ways. Examples:
 - cell size
 - cell morphology (shape)
 - metabolic strategies (physiology)
 - motility
 - mechanisms of cell division
 - pathogenicity and developmental biology

Introduction to Biodiversity

Microbial cell shapes



<http://dsc.discovery.com/news/2007/10/15/gallery/diatom-paint-540x540.jpg>



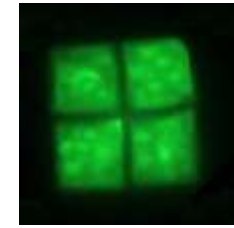
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<http://sis.biowaves.org/wp-content/uploads/2012/06/Streptomyces1.jpg>



http://en.wikipedia.org/wiki/File:Haloquadratum_walsbyi00.jpg



<http://www.astrobio.net/exclusive/3866/mutant-microbes-test-radiation-resistance>



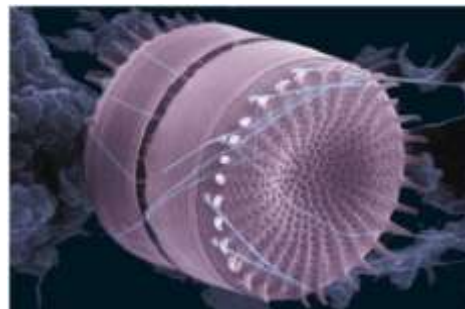
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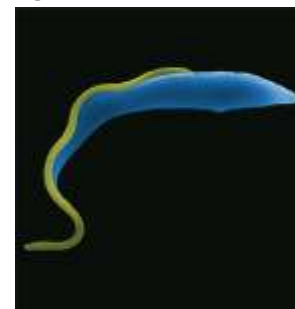
http://faculty.ccbcmd.edu/courses/bio141/labmanua/lab7/images/6937_Vibrio_monotrich.jpg



http://openi.nlm.nih.gov/imgs/rescaled512/3035206_sigs.492-f2.png



http://science.kennesaw.edu/~jdinrber/Bio2108/Lecture/LecBiodiversity/28_Labeled_Images/28_13FreshwaterDiatom-L.jpg



<http://ec.asm.org/content/10/7/F1.medium.gif>

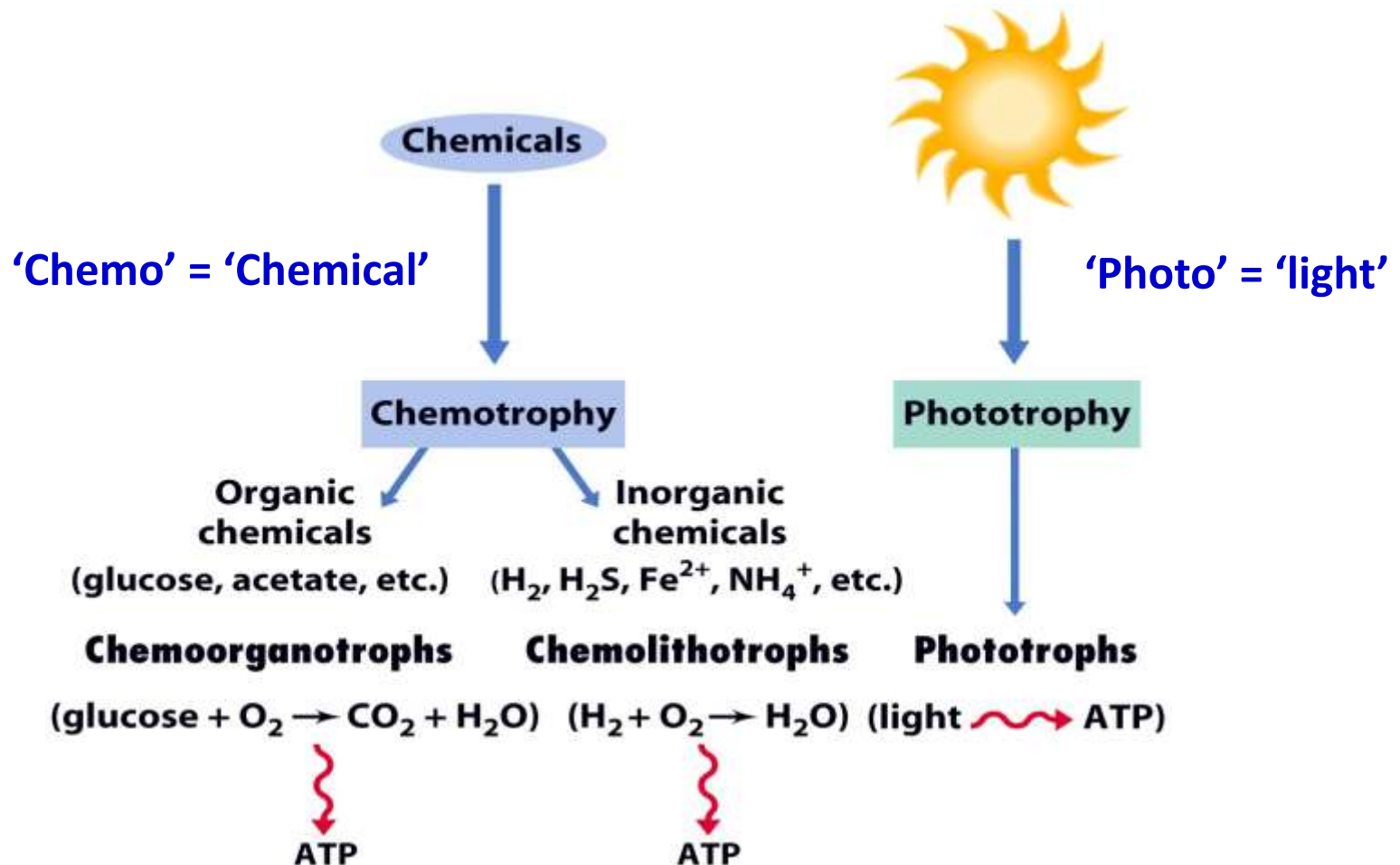
Introduction to Biodiversity

Physiological diversity of microbes

- All cells need energy and a means to trap it
- Cells obtain energy in nature in three ways:
 - 1) Oxidation of organic chemicals
 - 2) Oxidation of inorganic chemicals
 - 3) Conversion of light energy into chemical energy
- Oxidation = removal of electrons
Reduction = addition of electrons
- These processes are coupled during metabolism and electron transfer results in the production of high-energy compound **ATP**

Introduction to Biodiversity

Metabolic options to obtain energy I



Introduction to Biodiversity

Metabolic options to obtain energy II

- ***Chemoorganotrophs***: Microbes break down most of the natural and synthetic organic compounds to obtain the energy
- ***Chemolithotrophs***: Many prokaryotes obtain the energy from inorganic compounds. This metabolism is found only in prokaryotes (Bacteria and Archaea)
- ***Phototrophs***: Pigments allow them to use light as an energy source. They do not require chemicals; ATP is made from sunlight energy.
 - Oxygenic photosynthesis (O_2 is evolved e.g. cyanobacteria)
 - Anoxygenic photosynthesis (does not result in O_2 evolution e.g. purple and green bacteria)
- ***Aerobes and Anaerobes***: Some microbes extract energy only in the presence (aerobes) or absence of oxygen (anaerobes)

Introduction to Biodiversity

Autotrophs and heterotrophs

- All cells require carbon as a major nutrient
- Microbial cells are either autotrophs or heterotrophs
 - Autotrophs use **carbon dioxide** as their carbon source
 - Heterotrophs use **organic carbon**

Introduction to Biodiversity

Habitats and extreme environments

- Common microbial habitats:
 - Water
 - Soil
 - Animals
 - plants and man-made structures
- Some microbial environments are too extreme for humans; organisms inhabiting extreme environments are called **Extremophiles**

Introduction to Biodiversity

Examples of extremophiles

Table 2.1 Classes and examples of extremophiles^a

Extreme	Descriptive term	Genus/species	Domain	Habitat	Minimum	Optimum	Maximum
Temperature							
High	Hyperthermophile	<i>Pyrolobus fumarii</i>	Archaea	Hot, undersea hydrothermal vents	90°C	106°C	113°C ^b
Low	Psychrophile	<i>Polaromonas vacuolata</i>	Bacteria	Sea ice	0°C	4°C	12°C
pH							
Low	Acidophile	<i>Picrophilus oshimae</i>	Archaea	Acidic hot springs	−0.06	0.7 ^c	4
High	Alkaliphile	<i>Natronobacterium gregoryi</i>	Archaea	Soda lakes	8.5	10 ^d	12
Pressure	Barophile	<i>Moritella yayanosii</i> ^e	Bacteria	Deep ocean sediments	500 atm	700 atm	>1000 atm
Salt (NaCl)	Halophile	<i>Halobacterium salinarum</i>	Archaea	Salterns	15%	25%	32% (saturation)

^a In each category the organism listed is the current “record holder” for requiring a particular extreme condition for growth.

^b A newly isolated archaeon can apparently grow up to 121°C.

^c *P. oshimae* is also a thermophile, growing optimally at 60°C.

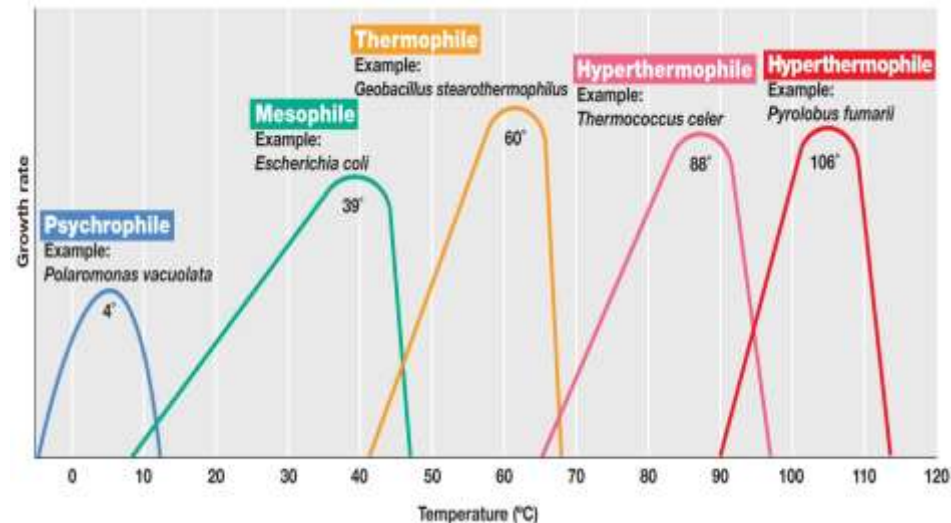
^d *N. gregoryi* is also an extreme halophile, growing optimally at 20% NaCl.

^e *Moritella yayanosii* is also a psychrophile, growing optimally at about 4°C.

Changes in Temperature

- Growth rate increases with temperature
- Proteins denature if temperature too high
- Microbial proteins adapted to temperature range:
 - Psychrophiles
 - Mesophiles
 - Thermophiles
 - Extreme thermophiles (Hyperthermophiles I & II)

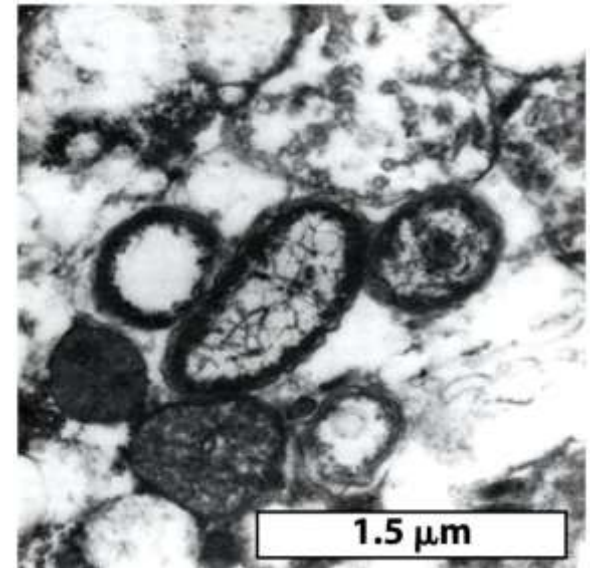
Relationship between temperature and growth rates of different groups of microbes



Variations in Pressure

- **Barophiles**
 - Adapted to high pressures
- **Barotolerant organisms**
 - Grow at high, but not very high pressure
- **Barosensitive organisms**
 - Die at high pressure
 - Most “typical” bacteria, all mammals

The barophile
Shewanella violacea



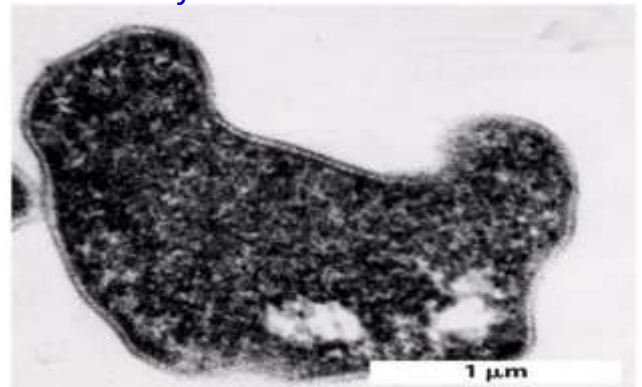
Changes in pH

- **Neutralophiles**
 - Grow at pH 5–8
 - Include bacteria in gut
- **Acidophiles**
 - Grow at pH 0–5
 - Some grow in stomach acid
 - Some in sulfuric acid springs
- **Alkalophiles**
 - Grow at pH 9–11
 - Found in soda lakes

Acidic spring in Mud Volcano area of
Yellowstone National Park, US



Thin-section of electron micrograph of
Sulfolobus acidocaldarius



Changes in Water Activity* - Halophiles

- **Halophiles (salt-loving organisms)**
- Require high concentration of NaCl
 - 2 – 4 M (10x seawater)
 - Live in salt seas

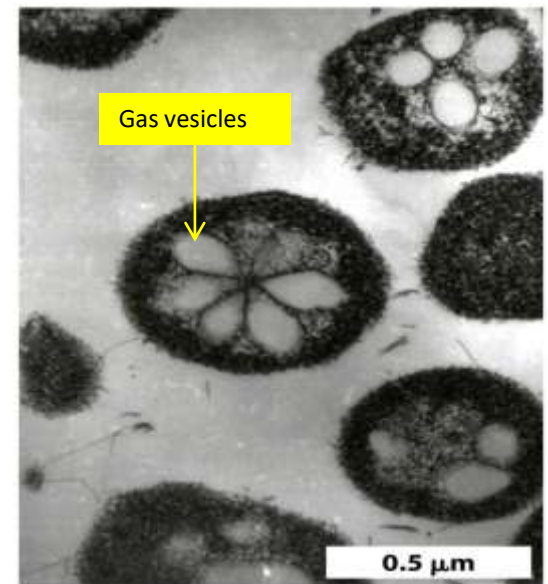
Halophilic salt flats in Naveda, US;
Pinkish colour produced by halophilic bacteria



Microbiology: An Evolving Science © 2009 W. W. Norton & Company, Inc.

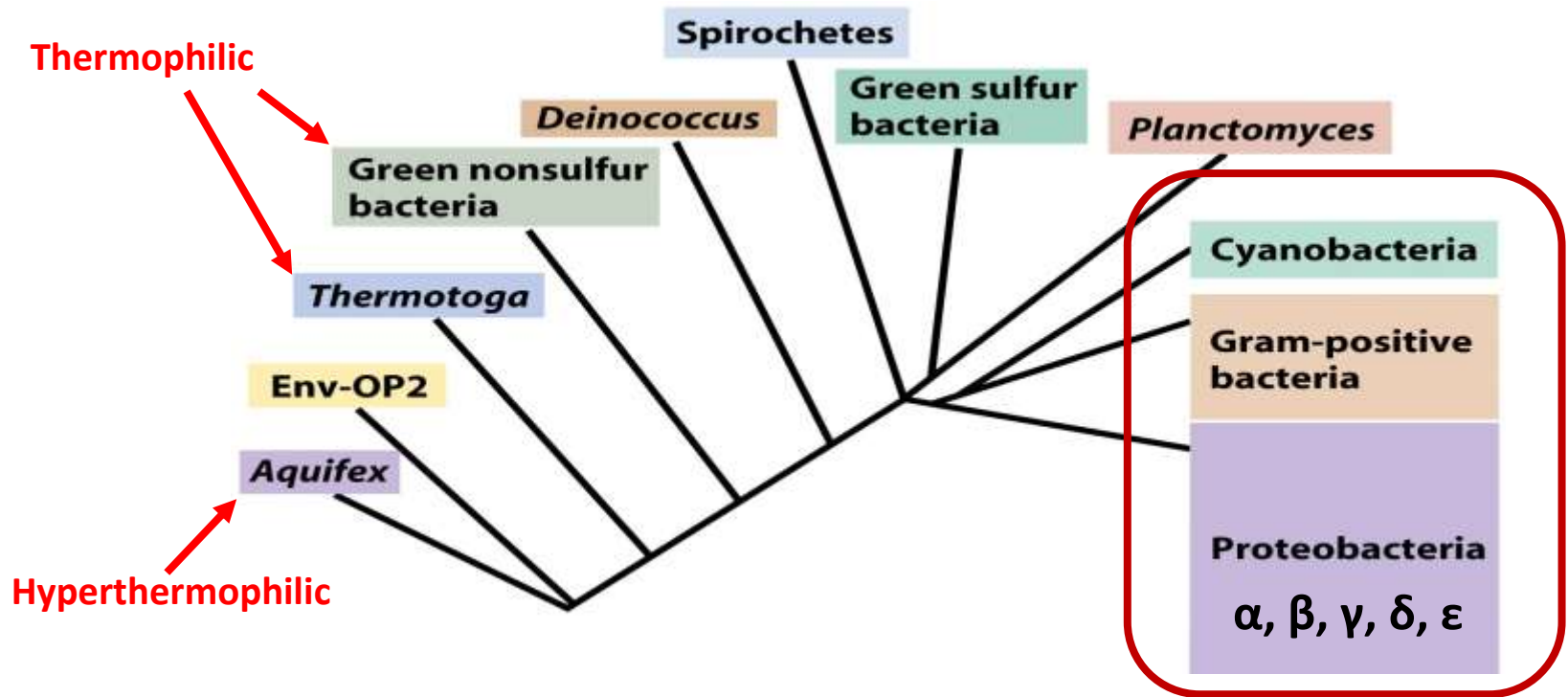
- Water activity or a_w is the partial vapor pressure of water in a substance divided by the standard state partial vapor pressure of water.
- Water activity is affected by the presence of solutes such as salts and sugars.

Cross section of the archaeon
Halobacterium sp.



Introduction to Biodiversity

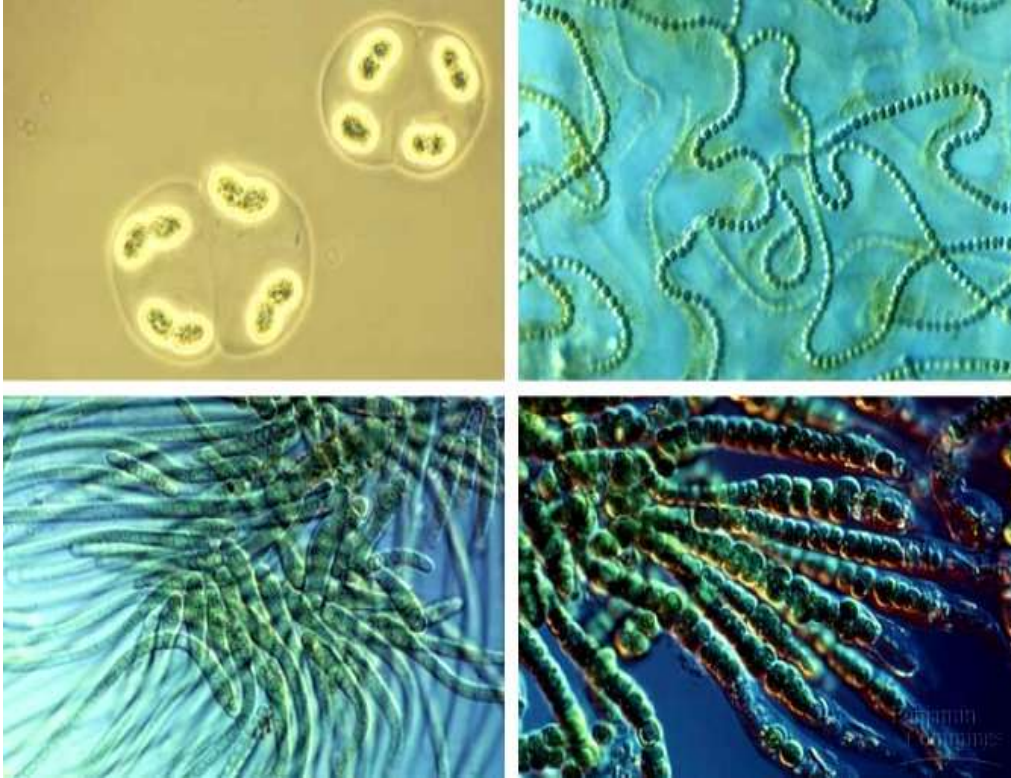
Phylogenetic diversity of Bacteria



- The **Proteobacteria** is the largest phylum of *Bacteria*.
Class: α -, β -, γ -, δ -, ϵ -proteobacteria.

Introduction to Biodiversity

Cyanobacteria

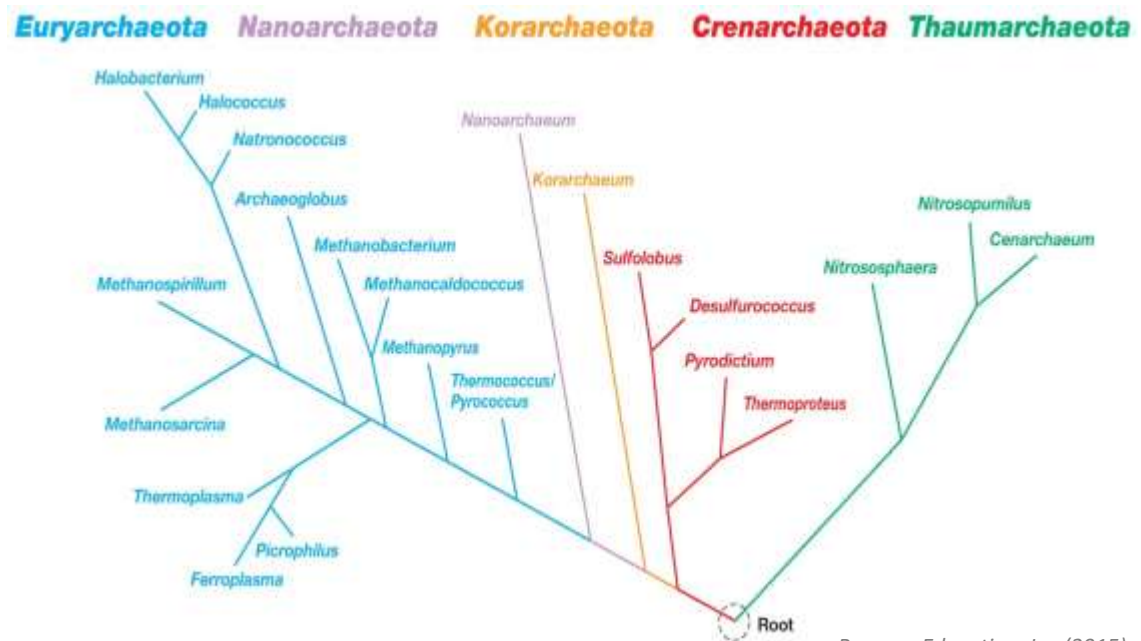


- The Cyanobacteria are phylogenetic relatives of Gram-positive bacteria and are (the first) oxygenic phototrophs
- Plants and animals developed when Earth had a more oxygen-rich environment
- Morphological diversity: Unicellular, colonial and filamentous forms

Introduction to Biodiversity

Diversity of the Archaea

- Archaea share many characteristics with both Bacteria and Eukarya
- Archaea are split into five major groups.
 - Euryarchaeota
 - Crenarchaeota
 - Thaumarchaeota
 - Korarchaeota
 - Nanoarchaeota



Introduction to Biodiversity

Diversity of the Eukaryotes

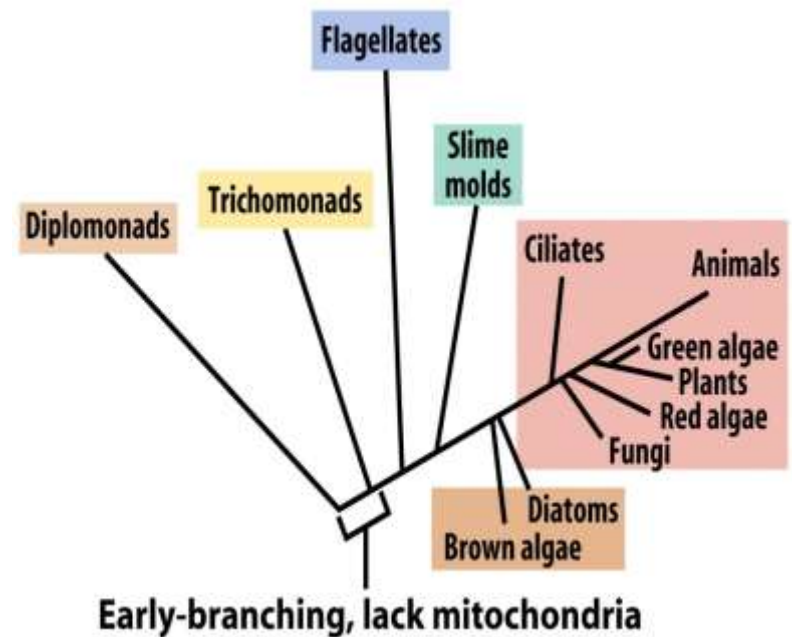
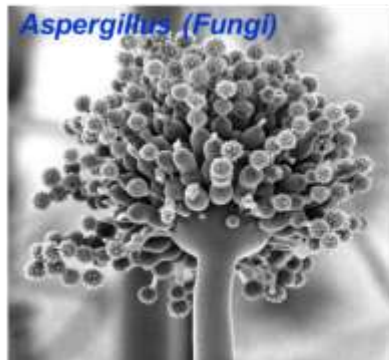
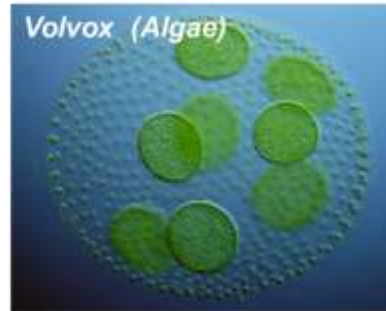
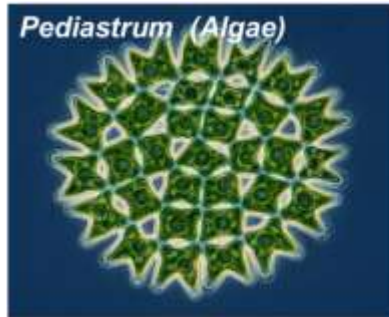


Figure 2-22 Brock Biology of Microorganisms 11/e
© 2006 Pearson Prentice Hall, Inc.

- Microbial eukaryotes are a diverse group that includes algae, protozoa, fungi, and slime molds
- Cells of algae and fungi have cell walls, whereas the protozoa do not

Introduction to Biodiversity

Eukaryotes - lichens



- Lichens – consists of symbiotic association of **fungus** (mycobiont) with a **photosynthetic partner** (photobiont) e.g. green alga or cyanobacterium
- They grow in most extreme environments (e.g. arctic tundra and hot deserts) and also abundant as epiphytes on leaves and branches in rain forests and temperate woodland

Introduction to Biodiversity

Eukaryotes - lichens

- **Key role in soil formation**
- **Grow on almost all surfaces,**
very slowly eroding them
- **Tolerate extreme environments**
temperature, lack of water
- **Sensitivity to pollutants –**
sulphur dioxide, heavy metals,
nitrogen compounds
- **Production of secondary metabolites**
Over 700 identified
Complex organic chemicals with
biological or pharmacological
properties. e.g. used in perfume industry
to enhance persistence of fragrance on
skin.

Lichens



<https://cn.bing.com/images>